LAB 5 – STATIC CODE ANALYSIS

Flight Reservation System

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Before — Pylint

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PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\before> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m pylint .\flig
ht_reservation_system.py
************* Module flight_reservation_system
flight_reservation_system.py:23:0: C0301: Line too long (132/100) (line-too-long)
flight_reservation_system.py:87:0: C0304: Final newline missing (missing-final-newline)
flight_reservation_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
flight_reservation_system.py:12:0: C0103: Constant name "total_revenue" doesn't conform to UPPER_CASE naming style (invalid-name)
flight_reservation_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:15:0: C0103: Function name "bookFlightSeat" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
flight_reservation_system.py:16:4: W0603: Using the global statement (global-statement)
flight_reservation_system.py:23:24: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
flight_reservation_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:27:0: C0103: Function name "cancelBooking" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:31:4: W0702: No exception type(s) specified (bare-except)
flight_reservation_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:35:0: C0103: Function name "checkSeatAvailability" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:37:7: C0121: Comparison 'flights.get(flight_no) == None' should be 'flights.get(flight_no) is None' (singleton-comparison)
flight_reservation_system.py:39:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
flight_reservation_system.py:39:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
flight_reservation_system.py:36:4: W0612: Unused variable 'airport_fee' (unused-variable)
flight_reservation_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:45:0: C0103: Function name "calculateBaggageFee" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:47:16: W0123: Use of eval (eval-used)
flight_reservation_system.py:46:4: W0612: Unused variable 'base_fare' (unused-variable)
flight_reservation_system.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:51:0: C0103: Function name "generatePassengerHash" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:55:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:55:0: C0103: Function name "generateBoardingPin" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:59:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:59:0: C0103: Function name "exportFlightManifest" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:60:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
flight_reservation_system.py:60:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
flight_reservation_system.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:66:0: C0103: Function name "getFlightSummary" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:66:21: W0621: Redefining name 'flights' from outer scope (line 6) (redefined-outer-name)
flight_reservation_system.py:68:8: W0612: Unused variable 'f' (unused-variable)
flight_reservation_system.py:74:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:4:0: W0611: Unused import math (unused-import)
```

Before — Flake8 and Bandit summary

```
--------------------------------------------------------------------


Code scanned:
        Total lines of code: 65
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 3
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 5
Files skipped (0):
```

```
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\before> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m flake8 .\flig
ht_reservation_system.py
.\flight_reservation_system.py:4:1: F401 'math' imported but unused
.\flight_reservation_system.py:7:80: E501 line too long (84 > 79 characters)
.\flight_reservation_system.py:8:80: E501 line too long (85 > 79 characters)
.\flight_reservation_system.py:9:80: E501 line too long (83 > 79 characters)
.\flight_reservation_system.py:23:80: E501 line too long (132 > 79 characters)
.\flight_reservation_system.py:31:5: E722 do not use bare 'except'
.\flight_reservation_system.py:36:5: F841 local variable 'airport_fee' is assigned to but never used
.\flight_reservation_system.py:37:31: E711 comparison to None should be 'if cond is None:'
.\flight_reservation_system.py:46:5: F841 local variable 'base_fare' is assigned to but never used
.\flight_reservation_system.py:87:7: W292 no newline at end of file
```

After — Program execution and Pylint

```
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" .\flight_reservat
ion_system.py
Is AI101 open: True
Baggage Fee: 67.5
Passenger Hash: 76161a689c7529645b02678017fed8b94a047e8aa458a82c50a23924f445ee07
Boarding PIN: 8540
Boarding manifest saved to flight_manifest.txt
Available flights: 3
```

After — Bandit

```
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m bandit .\fligh
t_reservation_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.10
Run started:2026-10-14 15:50:27.316258+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 88
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

After — Flake8

```
Files skipped (0):
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after> & "$HOME\Documents\L5_50_flight_reservation_system\venv\Scripts\python.exe" -m flake8 .\fligh
t_reservation_system.py
PS C:\Users\SHREYAS\Documents\SE_LAB_PES2UG24CS490\lab5\after>
```